

Indian Institute of Information Technology, Bhopal
Mini Test for 3rd Semester, 2025

Branch: CSE
Max Duration: 1 hour

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Subject Code: CSE-3005
MM=10

Attempting all the questions is mandatory.

Q. No.	Question	Marks
1. ✓	A modulated signal is given by, $s(t) = m_1(t) \cos(2\pi f_c t) + m_2(t) \sin(2\pi f_c t)$ where the baseband signal $m_1(t)$ and $m_2(t)$ have bandwidths of 10kHz and 15kHz, respectively. The bandwidth of the modulated signal, in kHz, is?	1
2. ✓	If a signal $f(t)$ has energy, the energy of the signal $f(2t)$ is equal to? With explanation. (a) E (b) $\frac{E}{2}$ (c) $2E$ (d) $4E$	2
3.	A message $m(t)$ band limited to the frequency f_m has a power of P_m . The power of the output signal in the figure is? 	2
4. ✓	The Fourier Transform of the signal $x(t) = e^{-3t^2}$ is of the following form, where A and B are constant. (a) Ae^{-Bt^2} (b) Ae^{-Bf^2} (c) $A + B f ^2$ (d) Ae^{-Bf}	2
5. ✓	For the modulated signal $x(t) = m(t) \cos(2\pi f_c t)$, the message signal $m(t) = 4 \cos(1000\pi t)$ and the carrier frequency f_c is 1 MHz. The signal $x(t)$ is passed through a demodulator, as shown in the figure below. Find the output $y(t)$ of the demodulator.	3

Handwritten notes and calculations:

For Q3: $P_m \cos^2 \theta / 4$

For Q5: $y(t) = 4 \cos(1000\pi t)$

For Q4: $\frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} |x(t)|^2 dt$